FLAT FINITE TYPE MAPS OF SPACES ACQUIRE FINITE PRESENTATION FROM A SCHEME-THEORETICALLY DENSE OPEN OF THE TARGET

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 8568-8580 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 8582-8588, the complete author proof (5 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. BORDERLINE for Chief: identical-proof pointer between two sibling labels (one published here, one queued B-side nearby); accepted under the written-substitution reading, flagged if pattern pointers are to be excluded.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The sibling dense-open lemma's published proof skeleton; More on Flatness, lemma flat-finite-type-finitely-presented-over-dense-open-X (the ring-theoretic engine variant).

Source reference. Original lemma 081P: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8588–8588. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0876: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8541–8551. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Let $V \subset Y$ be an open subspace. Assume*

(1) *f is locally of finite type and flat,*
(2) *$V \to Y$ is quasi-compact and scheme theoretically dense,*
(3) *$f|_{f^{-1}V} : f^{-1}V \to V$ is locally of finite presentation.*

Then f is of locally of finite presentation.

Proof. The proof is identical to the proof of Lemma 0876 except one uses More on Flatness, Lemma 081P. □

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.